Newton's Laws

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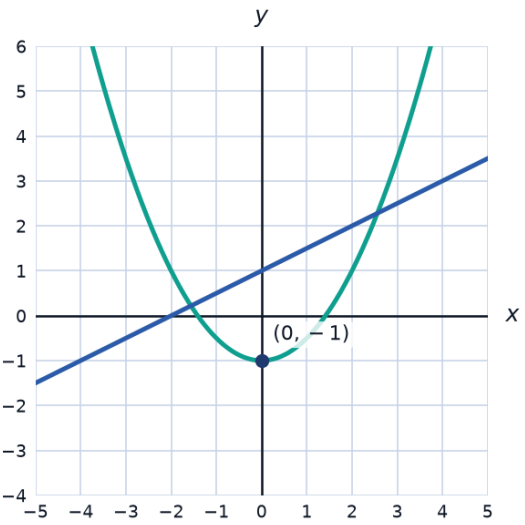
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Newton's Second Law ($F = ma$)

Particles of mass 11 kg and 16 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

2 [Foundational] • Target: Connected Particles over a Pulley

Particles of mass 9 kg and 22 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.



3 [Foundational] • Target: Motion on Rough Inclined Planes

Particles of mass 5 kg and 10 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

4 [Foundational] • Target: Connected Systems with Pulleys & Tables

Particles of mass 2 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

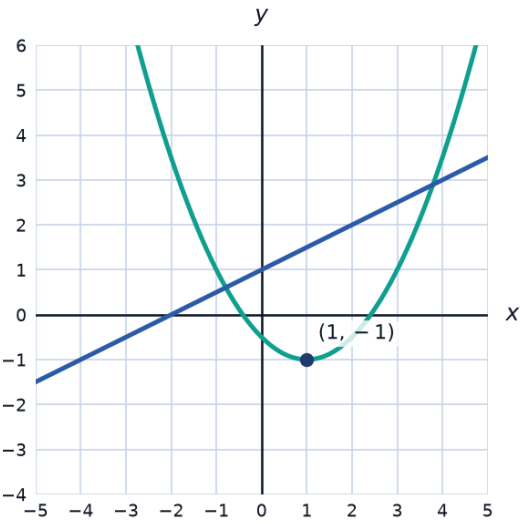
5 [Foundational] • Target: Newton's Second Law ($F = ma$)

Particles of mass 4 kg and 13 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

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6 [Foundational] • Target: Connected Particles over a Pulley

Particles of mass 7 kg and 17 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.



7 [Foundational] • Target: Motion on Rough Inclined Planes

Particles of mass 7 kg and 16 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Connected Systems with Pulleys & Tables

Particles of mass 6 kg and 7 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

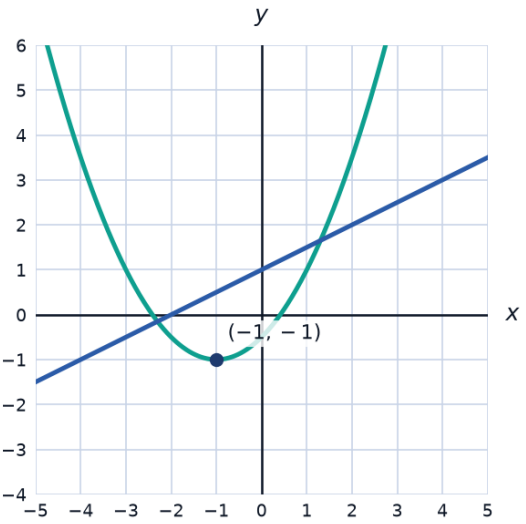
9 [Intermediate] • Target: Newton's Second Law ($F = ma$)

Particles of mass 4 kg and 11 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

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10 [Intermediate] • Target: Connected Particles over a Pulley

Particles of mass 2 kg and 5 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.



11 [Intermediate] • Target: Motion on Rough Inclined Planes

Particles of mass 5 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

12 [Intermediate] • Target: Connected Systems with Pulleys & Tables

Particles of mass 4 kg and 14 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

13 [Intermediate] • Target: Newton's Second Law ($F = ma$)

Particles of mass 1 kg and 7 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

14 [Intermediate] • Target: Connected Particles over a Pulley

Particles of mass 1 kg and 4 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the tension in the string.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

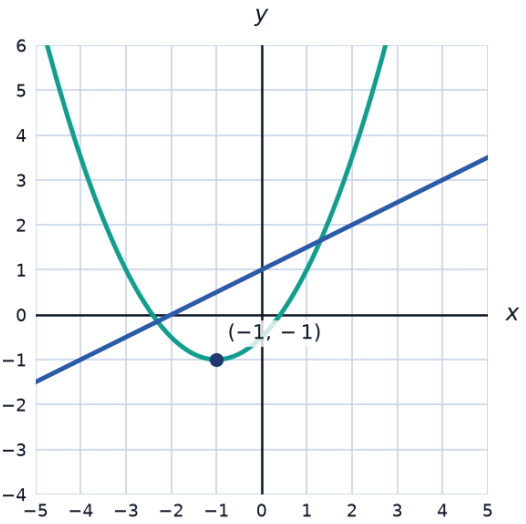
15 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 1.01 Indices, Surds & Quadratics

Particles of mass 11 kg and 26 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

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16 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 10.01 Series & Binomial Theorem

Particles of mass 12 kg and 14 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.



17 [Challenging] • Target: Newton's Second Law ($F = ma$) • Cross-Lesson: 11.01 Kinematics

Particles of mass 6 kg and 11 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

18 [Challenging] • Target: Connected Particles over a Pulley • Cross-Lesson: 12.01 Forces & Equilibrium

Particles of mass 8 kg and 10 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

19 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 14.01 Energy, Work & Power

Particles of mass 4 kg and 13 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

20 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 15.01 Representation of Data

Particles of mass 8 kg and 15 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Newton's Second Law ($F = ma$)

A constant resultant force of 36 N acts on a mass of 4 kg. What is the acceleration produced?

a. 9 m/s^2

b. 144 m/s^2

c. 4 m/s^2

d. 6 m/s^2

22 [Foundational] • Target: Newton's Second Law ($F = ma$)

A car of mass 1200 kg is traveling at 25 m/s when the brakes are applied, bringing it to rest in 50 m. Find the magnitude of the braking force.

a. 7500 N

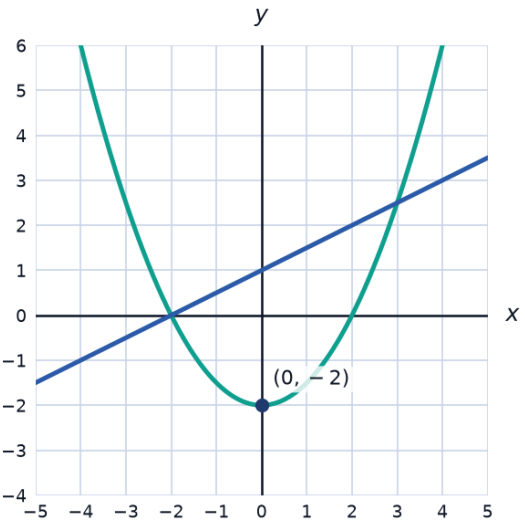
b. 6000 N

c. 9000 N

d. 15000 N

23 [Foundational] • Target: Newton's Second Law ($F = ma$)

An elevator of mass 800 kg accelerates upwards at $1.5\ m/s^2$. Taking $g = 10\ m/s^2$, find the tension in the cable.



a. 9200 N

b. 8000 N

c. 6800 N

d. 1200 N

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24 [Foundational] • Target: Connected Particles over a Pulley

Two masses of 3 kg and 2 kg are connected by a light inextensible string passing over a smooth fixed pulley (Atwood machine). Taking $g = 10$ m/s^2, find the acceleration of the system.

a. 2 m/s^2

b. 1 m/s^2

c. 4 m/s^2

d. 5 m/s^2

25 [Foundational] • Target: Connected Particles over a Pulley

For the system in the previous question, find the tension T in the string.

a. 24 N

b. 25 N

c. 20 N

d. 30 N

26 [Foundational] • Target: Connected Particles over a Pulley

What is the total force exerted by the string on the pulley in the previous system?

a. 48 N downwards

b. 24 N downwards

c. 50 N downwards

d. 40 N downwards

27 [Foundational] • Target: Motion on Rough Inclined Planes

A block of mass 5 kg is pulled up a smooth slope inclined at 30° to the horizontal by a force of 40 N acting parallel to the slope. Taking $g = 10$ m/s^2, find the acceleration of the block.

a. 3 m/s^2

b. 5 m/s^2

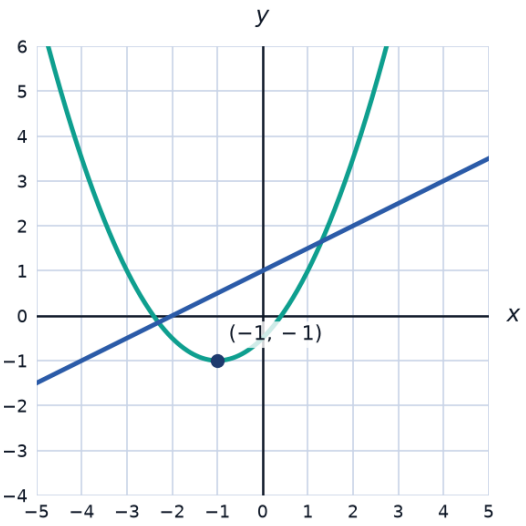
c. 2 m/s^2

d. 4 m/s^2

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28 [Foundational] • Target: Motion on Rough Inclined Planes

If the slope in the previous question is rough with $\mu = 0.2$, what is the new acceleration of the block?



a. $1.27\ \text{m/s}^2$

b. $3.00\ \text{m/s}^2$

c. $2.15\ \text{m/s}^2$

d. $0.85\ \text{m/s}^2$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Motion on Rough Inclined Planes

A sled of mass 20 kg slides down an incline of 25° at constant speed. What is the coefficient of kinetic friction μ?

a. $\mu = \tan(25^\circ) \approx 0.466$

b. 0.423

c. 0.906

d. 0.500

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30 [Intermediate] • Target: Connected Systems with Pulleys & Tables

A block A of mass 4 kg rests on a smooth horizontal table and is connected by a light string over a smooth pulley at the edge of the table to a hanging block B of mass 1 kg. Taking $g = 10$ m/s^2, find the acceleration.

a. 2 m/s^2

b. 2.5 m/s^2

c. 1.5 m/s^2

d. 10 m/s^2

31 [Intermediate] • Target: Connected Systems with Pulleys & Tables

If the table in the previous system has coefficient of friction $\mu = 0.15$, find the acceleration of the system.

a. 0.8 m/s^2

b. 1.2 m/s^2

c. 1.5 m/s^2

d. 0.5 m/s^2

32 [Intermediate] • Target: Connected Systems with Pulleys & Tables

A car of mass 1000 kg tows a trailer of mass 500 kg with acceleration 1.2 m/s^2. Resistance to motion is 300 N on the car and 200 N on the trailer. Find the tension in the towbar.

a. 800 N

b. 600 N

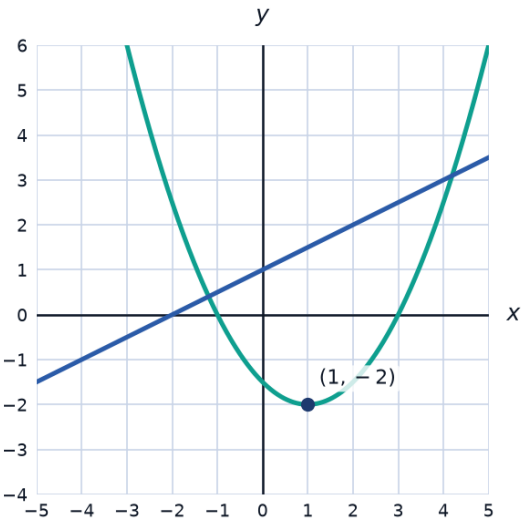
c. 1100 N

d. 1500 N

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33 **[Intermediate]** • **Target: Newton's Second Law ($F = ma$)**

Particles of mass 5 kg and 17 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.



a. 5.85

b. 5.35

c. 10.69

d. 4.85

34 **[Intermediate]** • **Target: Connected Particles over a Pulley**

Particles of mass 9 kg and 15 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 4.90

b. 2.45

c. 1.95

d. 2.95

35 **[Intermediate]** • **Target: Motion on Rough Inclined Planes**

Particles of mass 2 kg and 5 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 4.20

b. 8.40

c. 4.70

d. 3.70

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36 [Intermediate] • Target: Connected Systems with Pulleys & Tables

Particles of mass 1 kg and 3 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 4.90

b. 9.80

c. 5.40

d. 4.40

37 [Intermediate] • Target: Newton's Second Law ($F = ma$)

Particles of mass 3 kg and 4 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 0.90

b. 1.40

c. 1.90

d. 2.80

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Connected Particles over a Pulley • Cross-Lesson: 12.01 Forces & Equilibrium

Particles of mass 1 kg and 4 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 5.38

b. 11.76

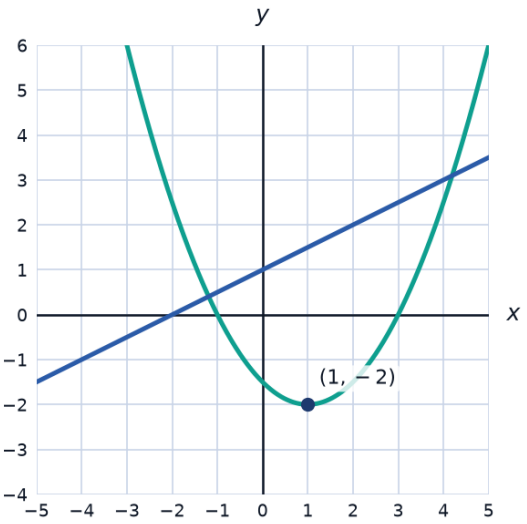
c. 5.88

d. 6.38

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39 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 14.01 Energy, Work & Power

Particles of mass 5 kg and 9 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.



a. 3.30

b. 2.30

c. 5.60

d. 2.80

40 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 15.01 Representation of Data

Particles of mass 2 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 5.38

b. 5.88

c. 6.38

d. 11.76

41 [Challenging] • Target: Newton's Second Law ($F = ma$) • Cross-Lesson: 16.01 Permutations & Combinations

Particles of mass 4 kg and 8 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 3.77

b. 6.53

c. 2.77

d. 3.27

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42 [Challenging] • Target: Connected Particles over a Pulley • Cross-Lesson: 17.01 Discrete Random Variables

Particles of mass 4 kg and 9 kg are connected by a light inextensible string over a smooth pulley, and released from rest. Taking $g = 9.8$ m/s^2, find the acceleration of the system.

a. 3.27

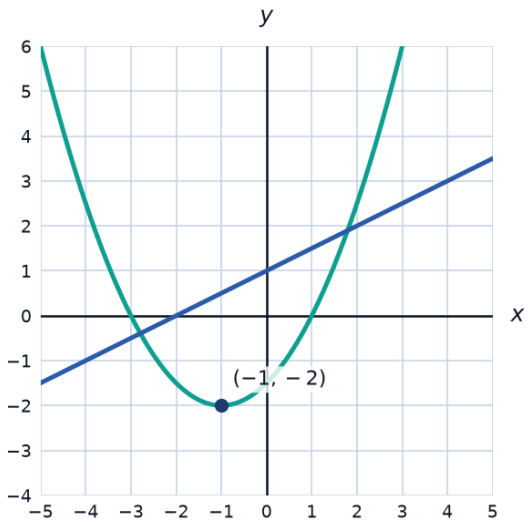
b. 3.77

c. 4.27

d. 7.54

43 [Challenging] • Target: Motion on Rough Inclined Planes • Cross-Lesson: 18.01 Binomial Distribution

A particle of mass 14 kg slides down a rough plane inclined at 60° to the horizontal, experiencing a constant friction force of 7 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.



a. 15.97

b. 8.49

c. 7.99

d. 7.49

44 [Challenging] • Target: Connected Systems with Pulleys & Tables • Cross-Lesson: 19.01 Normal Distribution

A particle of mass 12 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 9 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

a. 3.65

b. 4.15

c. 8.30

d. 4.65

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45 [Challenging] • Target: Newton's Second Law ($F = ma$) • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A particle of mass 12 kg slides down a rough plane inclined at 30° to the horizontal, experiencing a constant friction force of 3 N. Taking $g = 9.8$ m/s^2, find its acceleration down the slope.

a. 4.15

b. 4.65

c. 5.15

d. 9.30

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